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Remote operated gun breech

Abstract

A remote operated mechanism is provided for raising and lowering a seal in a gun loader breech via a transfer gear. The mechanism includes a servo motor, a pinion, a clutch and a worm gear. The servo motor axially rotates the pinion. The clutch controllably engages and disengages with the pinion. The worm gear couples the clutch to a shaft that engages the transfer gear. The clutch disengages the pinion from the motor in response to recoil from firing the gun. The worm gear includes a worm screw that axially turns with the pinion and a worm wheel that engages the worm screw to laterally spin a shaft connected to the transfer gear. The clutch includes a coupler for engaging the worm screw, a spur gear that connects the coupler to the pinion, and a spring that pushes the coupler to engage the worm screw prior to firing.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2877690	12/1958	Barr	89/43.01	F41A 25/02

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2327115	12/1998	GB	F41A 3/72

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The invention is a Continuation-in-Part, claims priority to and incorporates by reference in its entirety U.S. patent application Ser. No. 18/211,763 filed Jun. 20, 2023 and assigned Navy Case 210332.

BACKGROUND

(1) The invention relates generally to breech loaders for guns. In particular the invention relates to breech operation by remote control for a 105-mm gun used in cargo-carrying derivative aircraft.

(2) The 105 mm Gun Aircraft Unit (GAU) is an aerial gun system having a recoil mechanism and cradle integrated into a Trainable Gun Mount (TGM). The 105 mm GAU was designed specifically for integration into large aircraft (e.g., Lockheed AC-130H Spectre gunships) that perform ground support missions. The functional purpose of the GAU is to enable the aircraft to align to a target and fire artillery rounds at depressed angles while in flight.

(3) Prior to the exemplary 105 mm GAU, a special operation command for the U.S. Air Force relied on a modified cradle from the U.S. Army's M102 Howitzer along with the M37A1 recoil mechanism. The original design of the cradle and recoil mechanism was intended for use with a field artillery gun. This design has been adapted to aerial applications with select modifications, although various suboptimal constraints remain. This includes for example, the loader breech opens and closes via a manual crank.

SUMMARY

(4) Conventional breech loaders on aircraft guns yield disadvantages addressed by various

exemplary embodiments of the present invention. In particular, various exemplary embodiments provide a remote operated mechanism for raising and lowering a seal in a gun loader breech via a transfer gear. The mechanism includes a servo motor, a pinion, a clutch and a worm gear. The servo motor axially rotates the pinion. The clutch controllably engages and disengages with the pinion. The worm gear couples the clutch to a shaft that turns the transfer gear. The clutch disengages the pinion from the motor in response to recoil from firing the gun.

(5) In other various or additional embodiments, the worm gear includes a worm screw that axially turns with the pinion and a worm wheel that engages the worm screw to laterally spin a shaft connected to the transfer gear. The clutch includes a coupler for engaging the worm screw, a spur gear that connects the coupler to the pinion, and a spring that pushes the coupler to engage the worm screw prior to firing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and various other features and aspects of various exemplary embodiments will be readily understood with reference to the following detailed description taken in conjunction with the accompanying drawings, in which like or similar numbers are used throughout, and in which:

(2) FIG. 1 is an elevation and plan view of an AC-130 aircraft with an exemplary 105 mm Gun Aircraft Unit (GAU) with Trainable Gun Mount (TGM);

(3) FIG. 2 is an elevation view of the 105 mm GAU;

(4) FIGS. 3A and 3B are isometric views of the 105 mm GAU.

(5) FIG. 3C is an isometric detail view of the GAU aft section;

(6) FIG. 4 is an isometric assembly view of GAU recoil components;

(7) FIG. 5 is an isometric exploded view of the GAU;

(8) FIGS. 6A and 6B are isometric assembly views of a manually actuated breech;

(9) FIG. 6C is an isometric assembly view of the manual breech;

(10) FIGS. 7A and 7B are perspective assembly views of an automated breech;

(11) FIG. 8A is an isometric exploded view of the automated breech;

(12) FIGS. 8B, 8C and 8D are isometric assembly views of the breech;

(13) FIGS. 9A, 9B and 9C are perspective detail views of a coupling mechanism;

(14) FIGS. 10A and 10B are perspective detail views of a worm gear;

(15) FIG. 11 is an isometric exploded view of the worm gear;

(16) FIG. 12 is an isometric exploded view of clutch components;

(17) FIG. 13 is an isometric exploded view of coupler components;

(18) FIG. 14 is an isometric exploded view of the firing pawl; and

(19) FIG. 15 is an isometric exploded view of the manual firing lever.

DETAILED DESCRIPTION

(20) In the following detailed description of exemplary embodiments of the invention, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific exemplary embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments may be utilized, and logical, mechanical, and other changes may be made without departing from the spirit or scope of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present invention is defined only by the appended claims.

(21) The disclosure generally employs quantity units with the following abbreviations: length in meters (m) or inches ("), mass in grams (g) or pounds-mass (lbm), time in seconds(s), angles in degrees (°), and force in newtons (N) or pounds-force (lbf). Supplemental measures can be derived

from these, such as pressure in pounds-per-square-inch (psi), moment of inertia in gram-square-centimeters (kg-m.sup.2) and the like.

(22) FIG. 1 shows elevation views **100** of an AC-130 aircraft **110** as operated by the U.S. Air Force. An exemplary gun system **120** for a 105 mm Gun Aircraft Unit (**210** in FIG. 2) is disposed on a trainable gun mount atop the floor of the aircraft **110** to point laterally outward through a door hatch **130**. Section A-A points aft. Exemplary embodiments provide an exemplary operational kit for opening and closing the breech for the GAU **210** although such configurations can be applied to alternative gun systems.

(23) FIG. 2 shows an elevation view **200** of the 105 mm Gun Aircraft Unit (GAU) **210** for the gun system **120** along Section A-A disposed on a pallet **220** supported by rails **230** on the floor of the aircraft **110**. The 105 mm GAU **210** includes a gun tube **240** within a recoil sleeve **250** that connects to an annular flange **255**.

(24) Recoil can be mitigated by a brake assembly **260** and a pneumatic spring assembly **270** that flank the recoil sleeve **250** from above and below the gun tube **240**. Elevation actuator brackets **280** are disposed above the pallet **220** and flank the 105 mm GAU **210**. Fence arms **290** trailed by corresponding safety cages **295** straddle aft of the 105 mm GAU **210** to protect operators from recoil during gun fire. At the rear is an M102 manually operated breech for loading the 105 mm GAU **210**.

(25) FIG. 3A shows a perspective view **300** of the gun system **120**. A recoil cradle **310** includes the brake and spring assemblies **260** and **270** to mitigate recoil. A diffuser **320** attaches to the barrel **240** at its muzzle end. A modified M102 breech **330** terminates the 105 mm GAU **210** opposite the diffuser **320**. Floor trunnions **340** support the brackets **280** on the pallet **220**.

(26) An azimuth actuator **350** on an actuator bracket enables the 105 mm GAU **210** to yaw for engaging ground targets. An elevation actuator **360** disposed on the starboard bracket **280** pivots trunnions **370** to pitch a cradle tube **380** that constitutes the main structure of the cradle **310** in elevation. This enables the 105 mm GAU **210** to rise or depress in relation to the aircraft **110**.

(27) FIGS. 3B and 3C show detail perspective views **300** of the aft portion of the gun system **120**. The fence arms **290** include a fore mount flange **390** and an aft cage arm **395**.

(28) FIG. 4 shows an assembly view **400** of a 105 mm GAU **210** from the (proximal) muzzle end. A barrel recoiling assembly **410** is subdivided into two major sub-assemblies: the gun cradle **310** and the recoil mechanisms comprising the brake and spring assemblies **260** and **270** together with the gun tube **240** and recoil sleeve **250**.

(29) Identified components include breech nut **420**, hydraulic brake cylinder **430**, replenish cylinder **440**, pneumatic spring cylinder **450**, external counter-recoil buffers **460**, annular flange **255**, blast diffuser **320**, safety cage arms **290** and safety cage **295**. These cylinders **430**, **440** and **450** are hollow.

(30) FIG. 5 shows an isometric exploded view **500** of the 105 mm GAU **210**, including the recoiling assembly **310** and a manual breech assembly **510**. The recoil mechanism **310** includes the brake assembly **260** with brake cylinder **430** and replenish cylinder **440**, as well as the pneumatic spring assembly **270** with pneumatic spring cylinder **450** and a pair of counter-recoil buffers **460**. The manual breech assembly **510** includes the breech **330**, a crank mechanism **520**, a key **530** and a transfer cam **540**.

(31) The breech nut **420** attaches to the gun tube **240** and breech **330** to the recoil cradle **310**. The handle **520** opens the breech and key **530** locks the breech **330** to the recoil cradle **310**. Views **400** and **500** present components described more fully in the '763 application as ancillary components of the GAU **210**. By contrast, the present disclosure relates primarily to actuation of the breech **330**.

(32) In comparison with the '763 application, select components are assigned different drawing labels with the prior disclosure in parentheses. These include: annular flange **255** (**670**), fence arm **290** (**470**), safety cage **295** (**480**), diffuser **320** (**420**), breech **330** (**590**), azimuth actuator **350** (**550**),

elevation actuator **360 (560)**, cradle tube **380 (710)**, breech nut **420 (620)**, hydraulic brake cylinder **430 (630)**, replenish cylinder **440 (640)**, pneumatic spring cylinder **450 (650)**, external counter-recoil buffers **460 (660)**, handle **520 (1120)**, key **530 (1130)** and breech cam **540 (680)**.

(33) FIGS. **6A** and **6B** show assembly views **600** of closed and open dispositions of the breech **330** under manual operation. FIG. **6C** shows a supplemental assembly view **600** with the breech **330** closed. The breech **330** includes a housing **610** with a channel slot **615** and a sliding seal **620** that closes or opens the breech **330** by respectively raising or lowering within the housing **610**. A drive shaft **630** along the slot **615** operates a bevel gear transmission **640** within the transfer cam **540** to raise or lower the seal **620**.

(34) Lower and upper hinges **650** and **660** on the housing **610** support the shaft **630**, which connects to a turn arm **670** operated by a handle **675**. Turning the arm **670** clockwise from above pivots a coupler **680** that causes the seal **620** to rise or descend within the housing **610**. The transmission **640** includes a bottom bevel gear **690** responsive to the shaft **630** and an adjoining side bevel gear **695** that turns along the cam **540**.

(35) FIGS. **7A** and **7B** show assembly views **700** of respective closed and open dispositions of the breech **330** under exemplary motorized autoload operation powered by a linear servo motor **710** to raise and lower the seal **620** within the housing **610**. A worm gearbox assembly **720** transfers torque from the servo motor **710** to a drive shaft **730** supported by the lower hinge **660**. Additional components include a clutch assembly **740** equipped with a release handle **745**, a platform **750**, a mount cradle **760**, a coupler assembly **770**, a firing pawl **780** with a transmission **785** and a bracket **790**.

(36) FIG. **8A** shows an isometric exploded view **800** of the breech **330** and related motorized components. FIGS. **8B**, **8C** and **8D** show assembly views **800** of the breech **330** and fence **290**. The breech **330** includes the housing **610** with the channel slot **615**, the seal **620**, a shelf **810** for mounting the worm gearbox assembly **720** and a cavity opening **815** with external threads around the circumference.

(37) The gearbox assembly **720** includes a worm wheel **820** secured via a shim nut **825** to the shaft **730**, which includes a retaining ring **830**. The clutch assembly **740** includes shaft hole **840** for receiving a pinion **850** driven by the servo motor **710**.

(38) The mount flange **390** includes a face mount **860** to secure the cradle **760** and an attach flange **870** for securing the clutch assembly **740**. The fence **290** includes the cage arm **395** and the mount flange **390** connected together by a block **880** and separable by a release pin **885**. Auxiliary components include an override lever **890** and the manual firing pawl **780** with override transmission **785**.

(39) The clutch assembly **740** separates the servo motor **710** from the gearbox assembly **720** when the gun **110** fires, thereby pushing the breech **330** aft. Such recoil can damage sensitive components (e.g., the servo motor **710**), thereby necessitating clutch engagement to transfer rotational power to the shaft **730** while avoiding translational shock from firing.

(40) FIGS. **9A**, **9B**, and **9C** show perspective detail views **900** of components for the servo motor **710**, the firing pawl **780** and the clutch mechanism **770** that includes a spur gearbox **910**, a helical spring **920** and an override coupler **930**. In particular, FIG. **9B** shows the coupler **930** engaged with the spur gearbox **910**, while FIG. **9C** shows the coupler **930** disengaged, having moved forward away from the gearbox **910** as the override handle **745** rotates forward.

(41) The coupler **930** is always engaged to the spur gearbox **910**, views **700** and **900** show the (dis) engagement of the coupler **930** with the gearbox assembly **720** to enable manual actuation of the breech **330**. A platform **940** supports the firing pawl **780** atop the mount flange **390**. A fork brace **950** supports a bracket sleeve **960** and attaches to a block hub **970**. A cradle **980** atop the platform **940** supports the transmission **785**. The lever assembly **890** includes a handle **990**. Such manual interfaces **745**, **780** and **890** enable personnel to disengage equipment as necessary for operation.

(42) FIGS. **10A** and **10B** show perspective detail views **1000** of components for the worm gearbox

assembly **720**. A gearbox plate **1010** attaches to the breech housing **610** at the ledge **810** to form the gearbox structure and support an input gear shaft **1020**. A pair of fore and aft trunnions **1030** and **1040**. The worm screw **1020** includes a helical thread **1050** that interacts with the worm wheel **820**. A spring-loaded pawl clip **1060** secures the drive shaft **730** to the gearbox assembly **720**.

(43) FIG. **11** shows an isometric exploded view **1000** of components for the worm gearbox **720**. The plate **1010** includes a crank cavity **1110** for receiving the drive shaft **730** and peripheral mount holes **1115**. The worm screw **1020** includes the spiral thread **1050** onto an axial shaft **1120** flanked by fore (proximal) and aft (distal) end splines **1130** and **1140** and supported by bearings **1150** within respective trunnions **1030** and **1040**.

(44) The fore spline **1130** engages the coupler **930**. The aft spline **1140** provides manual drive input. The components are secured by the pawl clip **1060** together with a torsion spring **1155**, links **1160**, sleeve spacers **1170**, short bolts **1180** and corresponding washers **1185**, and long bolts **1190** and their washers **1195**. The torsion spring **1155** imposes tension to the pawl clip **1060** and the grooves **735** of the shaft **730**.

(45) The pin **885** acts to flip the pawl clip **1060**, which locks against ratchet grooves **735** of the drive shaft **730**. When the breech **330** is fully forward to be loaded the pin **885** holds the firing pawl **780** open and frees the shaft **730**. When the gun **110** recoils, the pin **885** releases the pawl **780**. The spring **920** forces the pawl **780** to engage the grooves **735** on the shaft **730** to prevent inadvertent opening of the breech **330**.

(46) FIG. **12** shows an isometric exploded view **1200** of the clutch assembly **740**. The spur gearbox **910** includes gear housing **1210** and cover **1215** with hole **840**. These contain a coupler gear **1220** flanked by bearings **1225**. The brace **950**, sleeve **960** and hub **970** comprise a load assembly **1230**. The sleeve **960** houses a shaft **1240**. The flange **950** attaches to the block hub **970** that receives the lever **745**, coupler **930**, compression spring **920**, drive shaft **1240**, release button **1245**, washer **1250** and retainer ring **1260**.

(47) These components are secured by #**1640-32** screws **1270**, washers **1275** and self-lock nuts **1280**, along with pin **1285** and #**1640-32** screw **1290** and washer **1295**. As the lever **745** pitches, the spring **920** compresses causing the coupler **930** and gear **1220** to withdraw from the fore spline **1130** of the worm gearbox **720**. The coupler gear **1220** turns the coupler **930** to release the sleeve **960** as the spring **920** applies tension between the brace **950** and the gearbox **910**.

(48) FIG. **13** shows an isometric exploded view **1300** of the manual lever **1325**. Components include pivot hub **970**, shaft **1310**, actuating fork **950**, bracket sleeve **960**, handle **745**, compression spring **1320**, plunger **1345**, bracket **1335**, linkage **1330**, pin **1350**, dowel **1355**, hex screw **1360**, hex screw **1365**, nut **1370**, cotter pin **1375**, cap screw **1380**, washer **1385**, dowel **1340**, dowel **1345**. Pivoting the handle **745** on the hub **970**, operates the actuating fork **950** on the shaft **730**.

(49) This assembly is used in conjunction with the spring loaded coupler **930** to disengage the motor **710** from the gearbox assembly **720** in the event the operator needs to manually actuate the breech **330**. While engaged, the motor brake locks the coupler **930** from rotating to prevent manually turning the worm screw **1050** via the hex spline end **1140**.

(50) FIG. **14** shows an isometric exploded view **1400** of firing pawl **780**. Components include grip **1410**, shaft hole **1415**, bracket **1420**, female threaded tube **1425**, shaft **1430**, male threaded follower **1440**, helical spring **1445**, shaft adapter **1450**, washer **1455**, slot spring **1460**, set screw **1465**, cotter pin **1470**, dowel **1475**. The transmission **785** comprises the bracket **1420** and the tube **1425**.

(51) As the grip **1410** turns the shaft **1430**, this assembly actuates by an electric solenoid (not shown) that pushes the grip **1410** into the transfer cam **540** on the breech **330** to fire the gun **110**. The shaft **1430** facilitates smooth operation to prevent binding and provide a manual override location in the event of power failure with the adapter **1450**. The spring **1445** returns the grip **1410** and transfer cam **540** to the ready fire position after actuation.

(52) FIG. **15** shows an isometric exploded view **1500** of the manual lever **890**. Components include

lever arm **1510**, pivot **1520**, actuating fork **1530**, **1535** dowel, **1540** pin, **1545** pin, cotter pin **1550**, **1560** plunger, hole **1570**, compression spring **1580**. This assembly engages with the shaft adapter **1450** of the firing pawl **780**. During normal remote operation the shaft adapter **1450** is free to actuate inside the fork **1530** without engaging the lever **890**.

(53) In the event of power or solenoid failure, the gun **110** can be fired by depressing the plunger **1560** and pivoting the lever **890**. This causes the fork **1530** to engage the shaft adapter **1450**, which pulls the firing pawl shaft **1430** connected to the grip **1410** to simulate the solenoid motion and fire the gun **110**.

(54) While certain features of the embodiments of the invention have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the embodiments.

Claims

1. A remote operated mechanism for raising and lowering a seal in a gun loader breech via a transfer gear, said mechanism comprising: a servo motor for axially rotating a pinion; a clutch for controllably engaging and disengaging with said pinion; and a worm gear for coupling said clutch to a shaft that turns the transfer gear, wherein said clutch disengages said pinion from said motor in response to recoil from firing the gun.
 2. The mechanism according to claim 1, wherein said worm gear further includes: a worm screw that axially turns with said pinion; and a worm wheel that engages said worm screw to laterally spin said shaft.
 3. The mechanism according to claim 2, wherein said worm screw mounts to a support platform that sets on a shelf on the breech.
 4. The mechanism according to claim 2, wherein said clutch includes: a coupler for engaging said worm screw; a spur gear that connects said coupler to said pinion; and a spring that pushes said coupler to engage said worm screw.
 5. The mechanism according to claim 1, wherein said motor and said clutch mount to a fence arm that connects to the gun.
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